SI Appendix:
Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention

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S1 Amino Acid Composition Profiles

We compared the per-position amino acid frequency profiles of the stored seed sequences and stochastic attention (SA)-generated sequences for the RNA recognition motif (RRM) family (PF00076, Fig. S1). The generated ensemble faithfully reproduces the conservation pattern of the family: strongly conserved positions (e.g., the aromatic residues characteristic of the ribonucleoprotein motifs RNP1 and RNP2) retain high frequency in the generated output, while variable positions display a similar breadth of amino acid usage. This agreement is consistent with the low Kullback–Leibler (KL) divergences reported in Table S1 and shows that the Boltzmann sampling procedure captures not only global amino acid composition but also the position-specific constraints encoded in the seed alignment.

S2 Generation Metrics

Table S1 reports the full generation metrics for all eight Pfam families across five sampling conditions: SA generation ($\beta \approx 2\beta^{*}$), SA retrieval ($\beta \approx 20\beta^{*}$), bootstrap resampling, Gaussian perturbation, and convex combination. For each condition, we report the KL divergence of amino acid composition, principal component analysis (PCA)-space cosine novelty, and nearest sequence identity to a stored pattern. Standard errors (SE) are computed over 30 independent Langevin chains of 5 samples each; KL standard errors are obtained via 1,000-fold bootstrap resampling. Bold entries indicate the best-performing method for each family and metric. The generation regime consistently achieves the best balance of compositional fidelity, novelty, and diversity across all eight families, while the convex combination baseline catastrophically degrades KL divergence in most families due to interpolation artifacts in PCA space.

S3 Scaling Study and Multi-Family Generalization

To characterize how generation quality degrades with decreasing family size, we subsampled the WW domain ($K{=}420$) at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition (Fig. S2). SA in the generation regime maintained low KL divergence across the entire range (0.019 ± 0.008 at $K{=}20$, improving to 0.010 ± 0.002 at $K{=}400$), while the convex combination baseline exceeded $\mathrm{KL}{=}0.8$ at every family size. Novelty increased with K (from 0.47 to 0.74), reflecting the richer landscape of a larger memory set, while nearest sequence identity decreased correspondingly (from 0.71 to 0.56). The empirical critical temperature β^{*} increased from 2.7 at $K{=}20$ to 5.5 at $K{=}400$, tracking the growth in PCA dimensionality. These results establish that SA generation is robust down to alignments as small as 20 sequences, though with reduced novelty reflecting the more constrained energy landscape.

The WW-only scaling study demonstrated robustness down to $K{=}20$, but a single family cannot establish generality. To test whether this robustness holds across families with different sequence lengths, conservation levels, and PCA dimensionalities, we subsampled three additional families to $K{=}20$ with five independent replicates each: SH3 ($K_{\mathrm{full}}{=}55$, $L{=}48$), Kunitz ($K_{\mathrm{full}}{=}99$, $L{=}53$), and zf-C2H2 ($K_{\mathrm{full}}{=}151$, $L{=}23$). All SA parameters

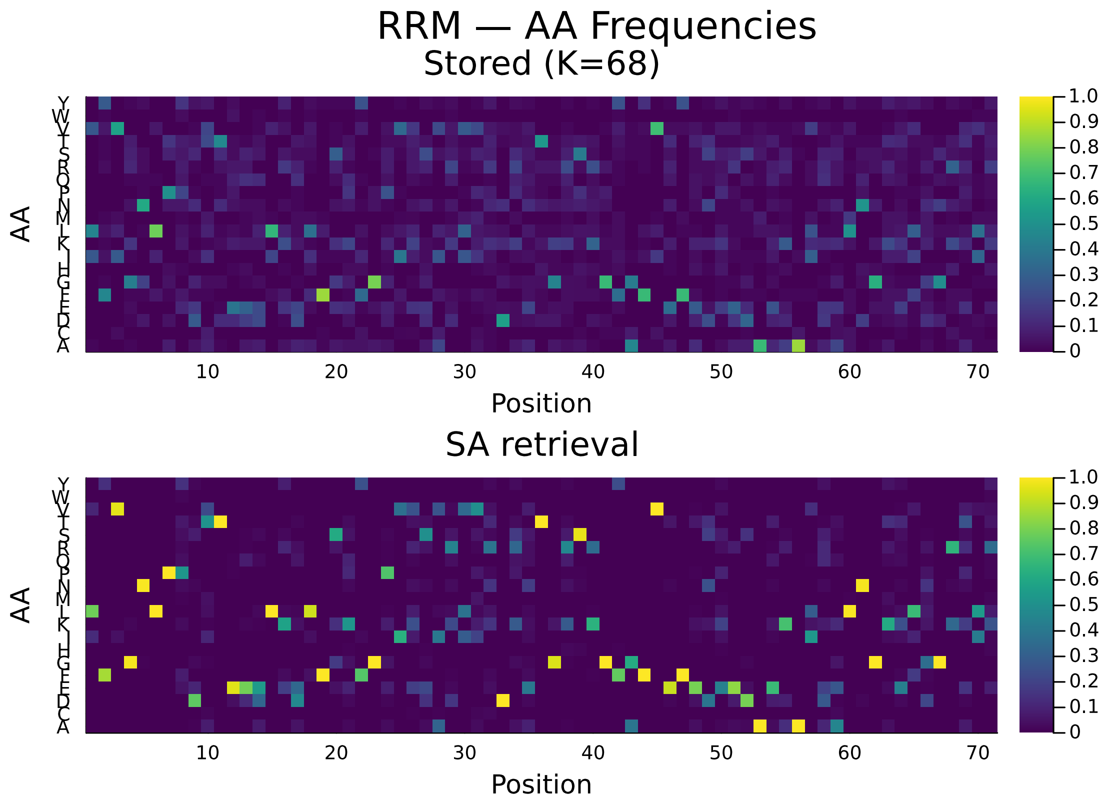

Figure S1: Per-position amino acid frequency profiles for the RRM family (PF00076). **Top:** stored seed sequences (K=68). **Bottom:** SA-generated sequences (S=150, retrieval regime). The generated ensemble reproduces the position-specific conservation pattern of the family, including strongly conserved positions and variable regions.

were identical to the main experiment; PCA and β^* were recomputed from scratch for each subsample. Table S2 reports the results alongside the original WW subsamples at K=20. SA maintained low KL divergence (0.013–0.026), substantial novelty (0.46–0.47), and moderate sequence identity (0.67–0.73) across all four families, with narrow standard deviations across the five replicates. The consistency of these metrics across families that differ in length by more than a factor of two (zf-C2H2: L=23 vs. Kunitz: L=53) confirms that the entropy inflection criterion successfully adapts the operating temperature to each family's geometry, and that the robustness observed in the WW scaling study is not family-specific.

S4 AlphaFold2 Cross-Validation

Figure S3 shows predicted local distance difference test (pLDDT) and template modeling (TM)-score results from both ESMFold and AlphaFold2 side by side across all eight families. The two predictors produce qualitatively identical patterns: SA-generated sequences achieve pLDDT well above the 70 threshold in every family under both predictors, and the family-to-family ordering of TM-scores is preserved. TM-scores were significantly higher for SA generation than for stored sequences in all eight families ($p < 0.05$; Cohen's d=0.47–1.73). Figure S4 quantifies the agreement directly: TM-scores are in near-perfect agreement across predictors (Pearson r=0.999), confirming that the structural ranking of conditions is not an artifact of ESMFold's architecture or training data. The pLDDT concordance is more moderate (r=0.434), reflecting the fact that each predictor has its own confidence calibration scale; AlphaFold2 pLDDT values are systematically higher than ESMFold values for the same sequences, consistent with known differences in the two models' confidence scoring. The near-perfect TM-score concordance is the key finding: it confirms that the structural conclusions drawn from ESMFold in the main text

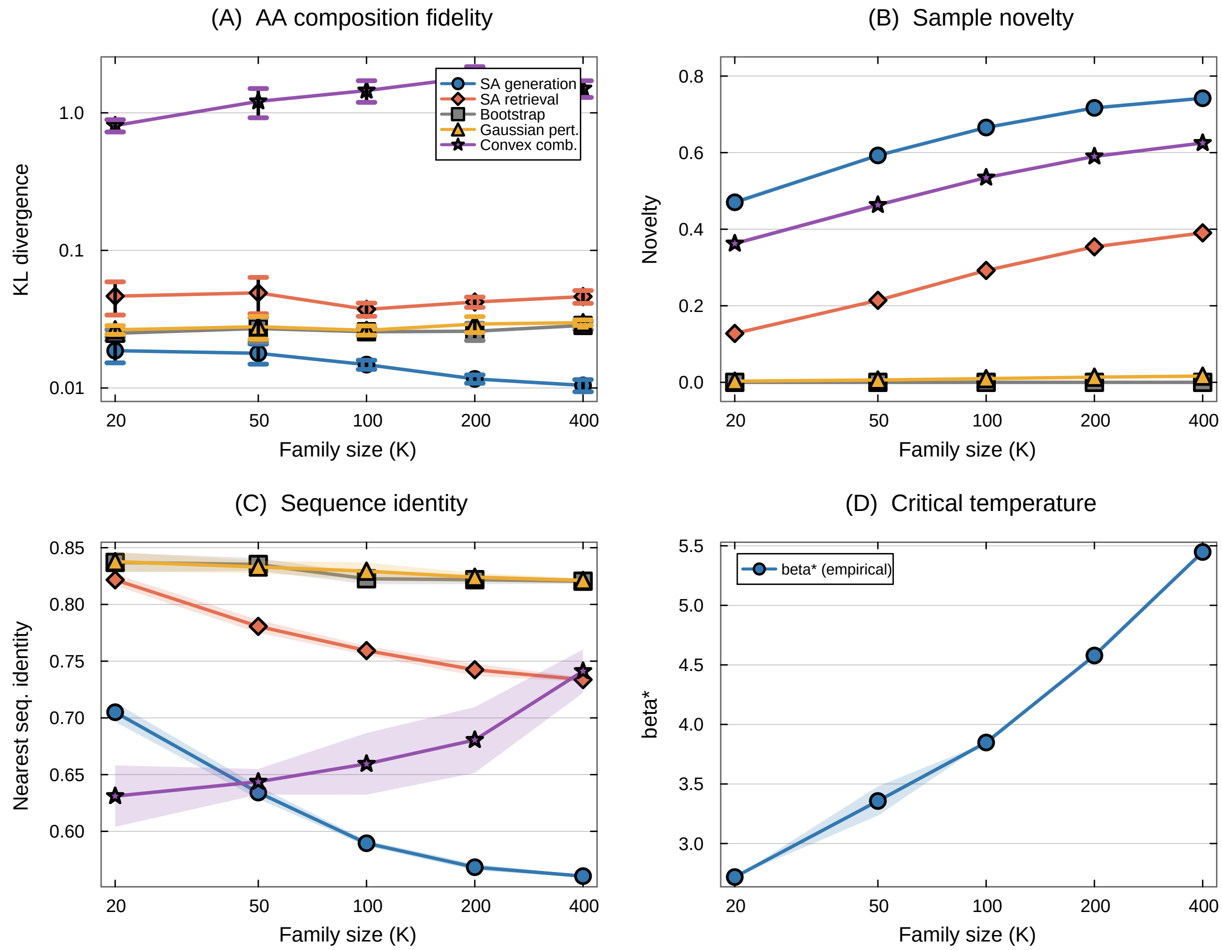

Figure S2: Scaling study: WW domain subsampled at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates. **(A)** KL divergence (log–log scale). SA generation maintains low KL across the entire range; convex combination degrades catastrophically. **(B)** Novelty increases with K for SA methods; baselines remain near zero. **(C)** Nearest sequence identity decreases with K, indicating the sampler explores further from stored patterns as more patterns shape the energy landscape. **(D)** The empirical critical temperature β^* increases with family size. Shaded regions and error bars show ± 1 SE.

are not predictor-specific.

S5 Sequence-Level Analysis

We aligned the five highest-confidence SA-generated Kunitz domain sequences against the family consensus to examine how the method handles the interplay between conservation and variation at the single-residue level (Fig. S5). All five sequences preserved every highly conserved position (11 of 11 positions with $> 90\%$ conservation in the seed alignment) and all six cysteines that form the three disulfide bonds essential to the Kunitz fold, while introducing 17–25 substitutions concentrated at variable positions. Per-position Shannon entropy correlated strongly between the SA-generated ensemble and the stored multiple sequence alignment (MSA; $r{=}0.968$), confirming that the method correctly identifies which positions tolerate variation and which do not. The concentration of substitutions at high-entropy positions is consistent with the Boltzmann sampling mechanism: the softmax attention weights distribute probability mass across stored patterns in proportion to their similarity to the current sample, so strongly conserved positions (where all stored patterns agree) are decoded to the consensus residue,

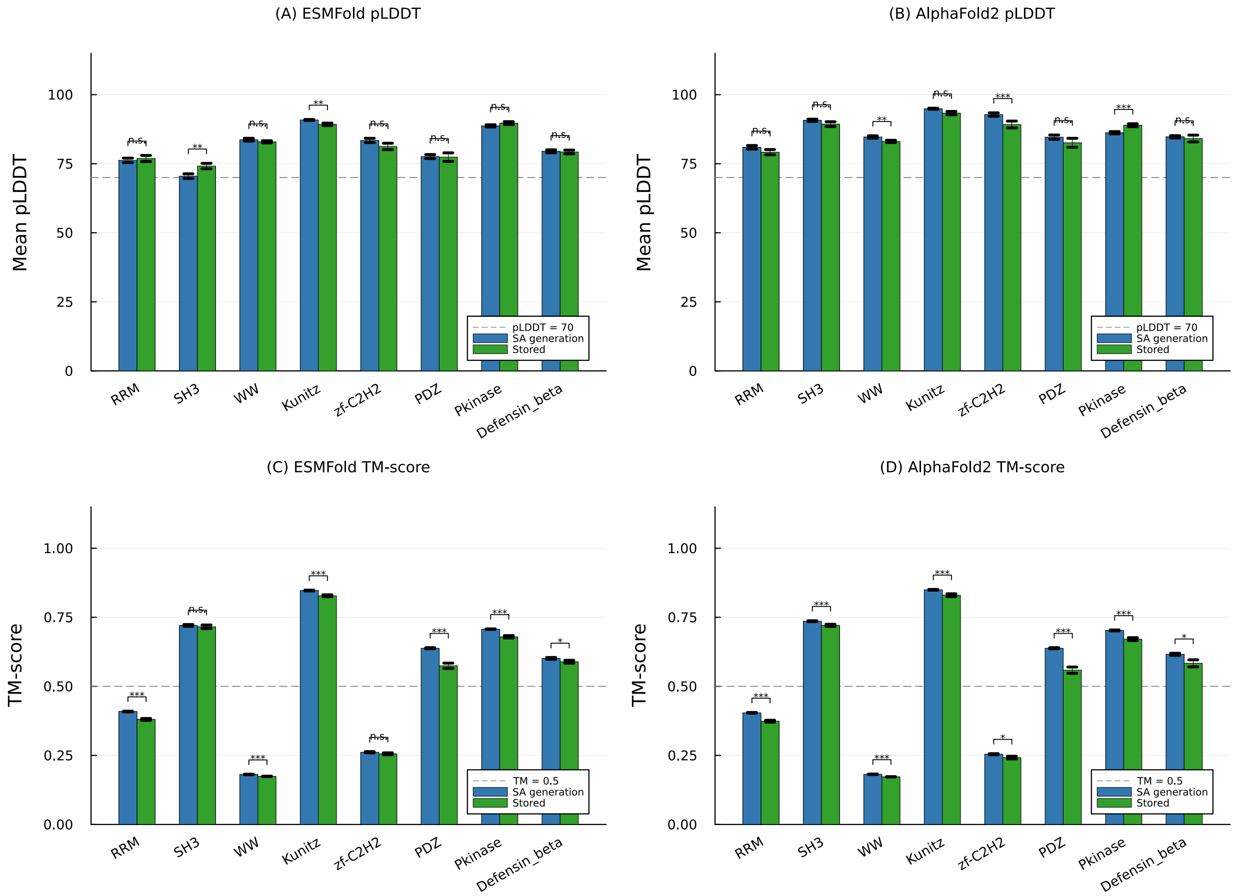

Figure S3: Structure validation results from ESMFold **(A, C)** and AlphaFold2 **(B, D)** across all eight Pfam families. **(A, B)** Mean pLDDT for SA-generated (blue) and stored (green) sequences; dashed line at pLDDT=70. **(C, D)** Mean TM-score to the family reference structure; dashed line at TM=0.5. Significance brackets show Wilcoxon rank-sum tests comparing SA generation to stored sequences. The two predictors produce qualitatively identical patterns across all families.

while variable positions (where stored patterns diverge) sample from the local distribution of amino acids.

S6 Pairwise Mutual Information Analysis

To assess whether SA-generated sequences preserve pairwise residue covariation, we computed mutual information (MI) matrices for the stored seed alignment and for sequences generated by each method. For each family, we constructed the $L \times L$ MI matrix, where position pair (i, j) has MI defined as $\mathrm{MI}(i, j) = \sum_{a,b} p(a, b) \log \frac{p(a,b)}{p_i(a)\, p_j(b)}$, with an additive pseudocount of 1.0 to regularize sparse counts. We then extracted the upper-triangle elements ($L(L-1)/2$ pairs) and computed Pearson and Spearman correlations between the stored MI vector and the MI vector of each generated ensemble. Table S3 reports the results across all eight families.

The central finding is that SA-generated sequences preserve the pairwise covariation structure of the stored alignment: Pearson r=0.53–0.92 in seven of eight families (mean r=0.72 excluding Pkinase), with SA consistently outperforming profile hidden Markov model (HMM) sequences (mean r=0.42), which model each position independently. The Pkinase family (K=37, L=262) was an outlier for all methods, consistent with the extreme data-to-parameter ratio limiting MI estimation quality in both the stored and generated ensembles. Methods that explicitly model pairwise couplings (Potts/plmDCA: r=0.75–0.89) or leverage pretraining on millions of MSAs

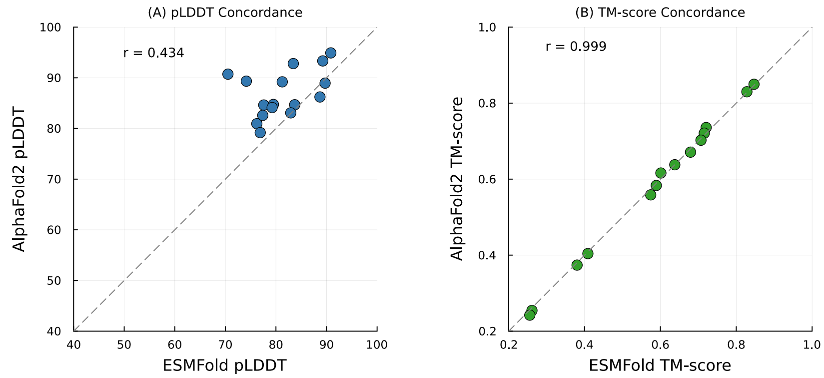

Figure S4: Concordance between ESMFold and AlphaFold2 structure predictions for SA-generated sequences across all eight families. Each point is one family-condition pair. **(A)** pLDDT concordance (r=0.434); AlphaFold2 pLDDT values are systematically higher, reflecting differences in confidence calibration. **(B)** TM-score concordance (r=0.999); points lie on the identity line, confirming that the two predictors rank conditions identically.

(EvoDiff: r=0.59–0.97) achieved higher MI correlations, as expected given their direct access to pairwise or higher-order statistics. The ordering HMM < SA < Potts ≤ EvoDiff is consistent with the information each method encodes: position-independent frequencies, the implicit all-order couplings of the Hopfield energy, explicitly fitted pairwise couplings, and deep coevolutionary priors from large-scale pretraining, respectively. Figure S7 shows this ordering across all eight families simultaneously, and Fig. S6 shows MI scatter plots for representative families, with structural contact pairs highlighted to confirm that SA preserves covariation at structurally relevant positions.

S7 Independent Fitness Validation via ESM2

As an independent assessment of whether SA-generated sequences resemble natural proteins, we scored all generated and stored sequences using ESM2-650M [1], a protein language model pretrained on millions of UniRef50 sequences that is entirely independent of the SA framework. For each sequence, we computed the pseudo-perplexity via masked marginal scoring: each position was masked in turn, the model predicted a distribution over amino acids, and the log-probability of the true residue was recorded. The pseudo-perplexity (lower is more protein-like) is $\exp(-\frac{1}{L}\sum_{\ell=1}^{L}\log P(a_\ell \mid \mathbf{a}_{\setminus\ell}))$, where a_ℓ is the amino acid at position ℓ and $\mathbf{a}_{\setminus\ell}$ denotes all other positions. Wilcoxon rank-sum tests with Cohen's d effect sizes compared SA-generated sequences (n=150) to stored natural sequences (n=37–50) for each family (Table S4).

SA-generated sequences were statistically indistinguishable from natural family members in four of eight families (Wilcoxon $p > 0.05$; RRM, zf-C2H2, Pkinase, Defensin_beta: $p > 0.29$). In the Kunitz domain, SA generation achieved significantly *lower* (better) pseudo-perplexity than stored sequences (4.47 vs. 4.98, $p < 0.001$, d=−0.74). Three families showed modestly higher pseudo-perplexity for SA generation (SH3: d=0.60; WW: d=1.03; PDZ: d=0.60), though all SA-generated sequences remained well within the range of natural protein sequences. The overall mean pseudo-perplexity across all families was 5.71 for SA-generated sequences versus 5.58 for stored sequences, confirming that an independent protein language model trained on billions of sequences considers SA-generated sequences to be comparably protein-like to natural family members.

S8 Deep Mutational Scanning Validation

To assess whether SA-generated substitutions correspond to experimentally tolerated mutations, we cross-referenced the amino acid changes introduced by SA generation against published deep mutational scanning (DMS) datasets for two families: the PDZ domain (DLG4/PSD-95 PDZ3; McLaughlin et al. [2]) and the SH3 domain (GRB2 SH3; Faure et al. [3]). For each family, we identified the DMS reference protein in the Pfam seed

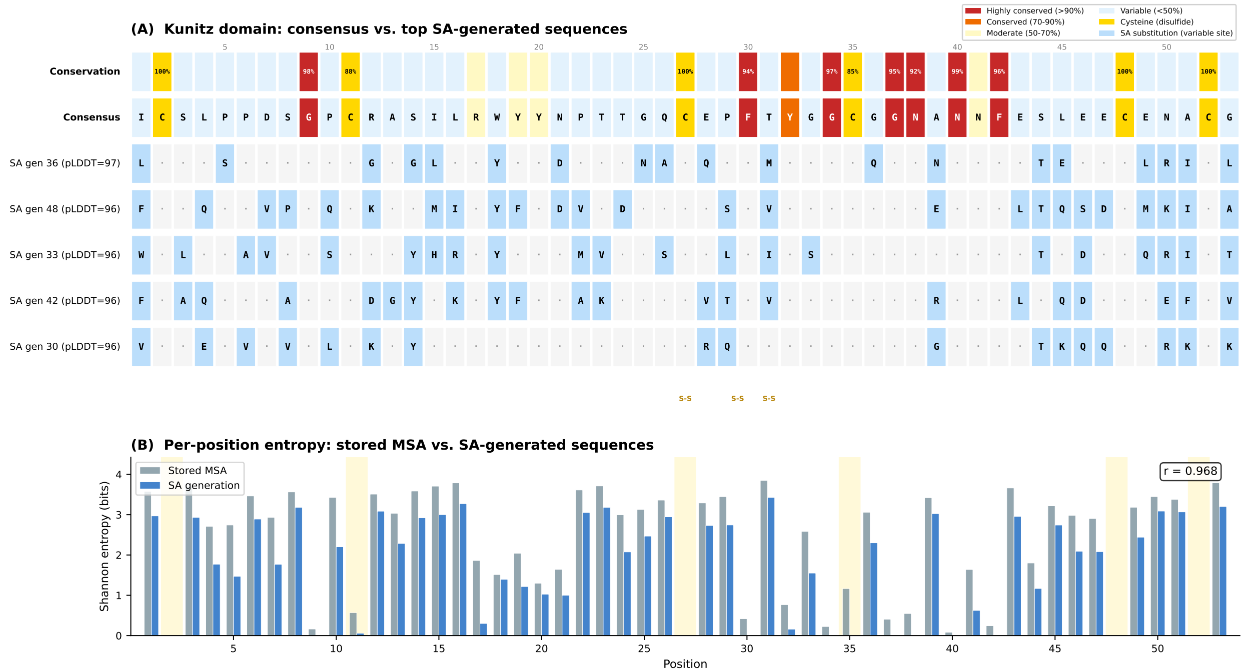

Figure S5: Sequence-level analysis of SA-generated Kunitz domain sequences. **(A)** Alignment colored by conservation level. **(B)** Per-position Shannon entropy correlation between the SA-generated ensemble and the stored seed alignment (r=0.968).

alignment, mapped alignment positions to DMS coordinates, and classified each SA-generated substitution relative to the family consensus as experimentally tolerated ($\mathrm{DMS\_score\_bin}{=}1$) or deleterious ($\mathrm{DMS\_score\_bin}{=}0$) based on the published fitness data (Table S5).

SA-generated substitutions were strongly enriched for experimentally tolerated mutations in both families. In the PDZ domain, 91.7% of the $4{,}313$ matched substitutions were classified as tolerated, compared to an overall tolerance rate of 77.5% in the DMS dataset ($1.18\times$ enrichment, binomial $p < 10^{-135}$). The continuous DMS fitness scores of SA-generated substitutions were also significantly higher than those of random single mutations (Mann-Whitney $p < 10^{-21}$, Cohen's d=0.42). In the SH3 domain, the enrichment was even stronger: 90.4% of $2{,}396$ matched substitutions were tolerated versus 67.5% overall ($1.34\times$ enrichment, binomial $p < 10^{-156}$), with a larger effect size on continuous fitness scores (d=0.70, $p < 10^{-55}$). These results demonstrate that the Boltzmann sampling procedure preferentially generates substitutions at positions and to residues that have been experimentally validated as functional, without any explicit fitness objective or training signal.

S9 Potts Model (plmDCA) Baseline

To assess whether pairwise coevolutionary models provide a stronger baseline than profile HMMs, we fitted a Potts model via pseudo-likelihood maximization (plmDCA) [4] to the SH3 and Kunitz seed alignments and generated 150 sequences per family by Gibbs sampling (500 sweeps per chain, initialized from random stored sequences). The Potts model captures pairwise couplings $J_{ij}(a,b)$ between all position pairs in addition to single-site fields $h_i(a)$, giving it substantially more parameters than a profile HMM. For SH3 (K=55, L=48), the model has approximately $498{,}000$ parameters for 55 training sequences; for Kunitz (K=99, L=53), approximately $609{,}000$ parameters for 99 sequences. Despite this extreme over-parameterization, L2 regularization (λ_J=λ_h=0.01) yielded well-behaved generation (Table S6).

The Potts model achieved comparable or slightly better amino acid composition fidelity than SA (KL divergence 0.025 vs. 0.036 for SH3; 0.010 vs. 0.013 for Kunitz) while producing sequences with modestly lower novelty (0.37 vs. 0.47 for SH3; 0.54 vs. 0.56 for Kunitz). Both methods occupy the same intermediate regime between

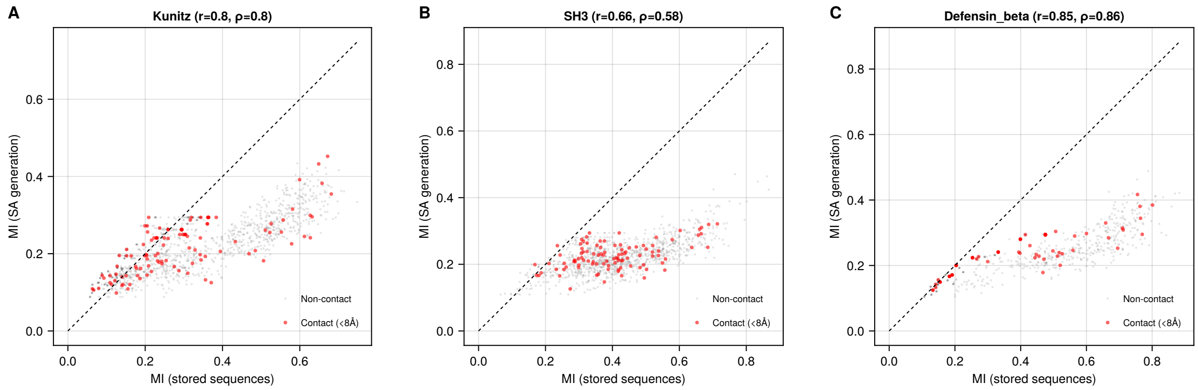

Figure S6: Pairwise MI scatter plots for representative families. Each point represents one position pair (i, j); the x-axis is MI from the stored seed alignment and the y-axis is MI from SA-generated sequences. Red points indicate structurally contacting residue pairs (Cα–Cα distance <8 Å, sequence separation ≥ 5 residues). Pearson r and Spearman ρ are shown in each panel.

profile HMMs (which drift far from the family at 23–29% identity) and bootstrap resampling (which has zero novelty). The critical distinction is computational: the Potts model required fitting $\sim 500{,}000$ parameters via L-BFGS-B optimization at each of L positions, followed by 500 Gibbs sweeps per sample (total wall time: ~ 100 s per family), while SA required no fitting and generated sequences in seconds. The similar output quality supports the conclusion that the information captured by pairwise couplings is also present in the Hopfield energy landscape, which implicitly encodes all-order correlations through the softmax attention operation.

S10 Sampling Diagnostics

To characterize the mixing properties of the Langevin sampler and justify the burn-in and thinning parameters used throughout the paper, we computed autocorrelation functions and effective sample sizes (ESS) from the Hopfield energy trace $E(\boldsymbol{\xi}_t)$ along each chain. For each family, we ran 10 independent unadjusted Langevin algorithm (ULA) and Metropolis-adjusted Langevin algorithm (MALA) chains of $T{=}5{,}000$ steps at the generation temperature $\beta_{\text{gen}} \approx 2\beta^*$, recording the full trajectory. We computed the normalized autocorrelation function of the post-burn-in energy trace (iterations $2{,}001$–$5{,}000$), estimated the integrated autocorrelation time τ_{int} using the initial positive sequence estimator with a cutoff at $\rho(\tau) < 0.05$, and derived the effective sample size as $\text{ESS} = n/\tau_{\text{int}}$ where $n{=}3{,}000$ is the number of post-burn-in samples.

Table S7 reports the results across all eight families. The integrated autocorrelation time ranged from 76 to 122 iterations for ULA and from 82 to 131 for MALA, yielding 32–42 effective samples per chain (ULA) and 27–39 per chain (MALA). The thinning interval of 100 is thus well matched to the autocorrelation time ($\text{thin}/\tau_{\text{int}} = 0.8$–$1.3$), ensuring that retained samples are approximately independent. Across 30 chains, the total effective sample size is approximately $30 \times 35 \approx 1{,}050$ independent draws from the Boltzmann distribution, from which we retain 150 decoded sequences (5 per chain). The burn-in of $2{,}000$ iterations provides a 3.2–$6.6\times$ margin over the empirical convergence point (304–617 iterations), confirming that the chains have reached stationarity before sampling begins. The MALA acceptance rates (99.6–99.8%) confirm that the ULA discretization bias is negligible at step size $\alpha{=}0.01$: the Metropolis correction rejects fewer than 0.4% of proposals, indicating that the unadjusted and adjusted samplers explore essentially the same distribution.

S11 Initialization Sensitivity

The multi-chain protocol described in Materials and Methods initializes each Langevin chain near a randomly selected stored pattern ($\boldsymbol{\xi}_0 = \mathbf{m}_k + \sigma_{\text{init}}\boldsymbol{\epsilon}$, $\sigma_{\text{init}} = 0.01$). To test whether this choice affects the generated ensemble, we repeated the full SA generation pipeline with random initialization on the unit sphere: each chain

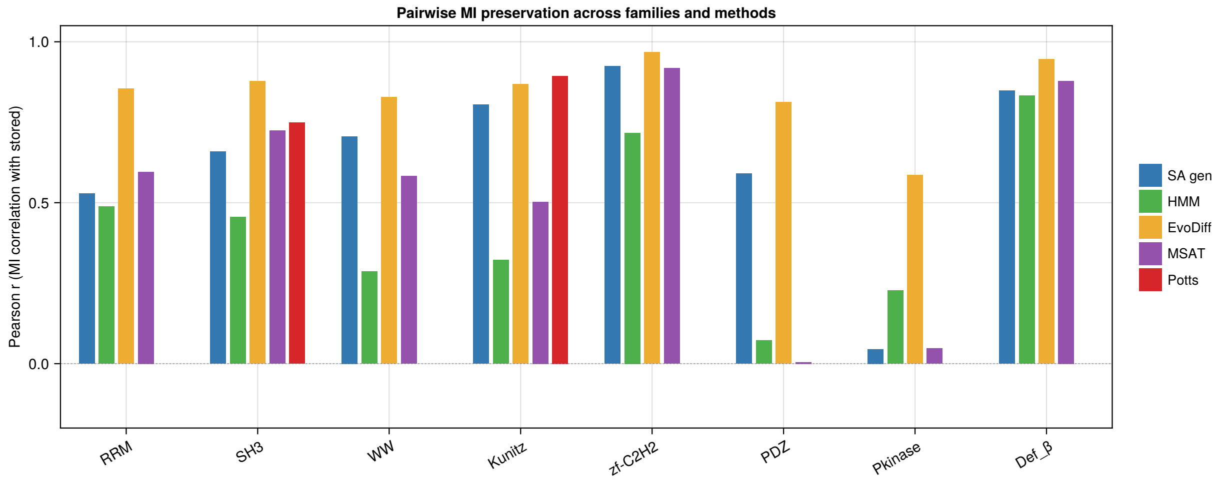

Figure S7: Pairwise MI preservation across all eight Pfam families. Each bar shows the Pearson correlation between the upper-triangle MI vector of the stored seed alignment and the MI vector of sequences generated by the indicated method. The consistent ordering HMM $<$ SA gen $<$ Potts $\leq$ EvoDiff reflects the information hierarchy of each model. Potts results are shown only for SH3 and Kunitz (red diamonds). Pkinase is a consistent outlier for all methods owing to its extreme data-to-position ratio (K=37, L=262).

was started from $\boldsymbol{\xi}_0 = \mathbf{z}/\|\mathbf{z}\|$ with $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, placing the initial point at a random location unrelated to any stored pattern. All other parameters were held fixed.

The results were unambiguous: random sphere initialization produced output indistinguishable from stored-pattern initialization across all eight families (Table S8). The maximum absolute difference in KL divergence was 0.0003 (RRM), and all other families showed differences below 10^{-4} in every metric. This insensitivity is expected from the theory: at $\beta_{\text{gen}} \approx 2\beta^*$, the energy landscape has well-defined basins near stored patterns, and the dissipative drift term $-\alpha\nabla E(\boldsymbol{\xi})$ rapidly pulls any initial condition toward the nearest basin. The 2,000-step burn-in (at step size α=0.01, corresponding to 20 effective time units) is more than sufficient for the chains to reach stationarity, erasing all memory of the starting point. The result confirms that stored-pattern initialization is a computational convenience (reducing burn-in time) rather than a requirement, and that the generated ensemble reflects the Boltzmann distribution rather than an artifact of the initialization strategy.

S12 Near-Duplicate Analysis

A key concern for any generative model operating on small datasets is whether the generated sequences are trivially close to stored patterns, effectively copying inputs with minor perturbations rather than producing genuinely novel outputs. To quantify the degree of near-duplication, we computed the full pairwise sequence identity between every SA-generated sequence (S=150) and every stored sequence (K) for each family, yielding a $150 \times K$ identity matrix. For each generated sequence, we recorded the maximum identity to any stored sequence (nearest neighbor) and the minimum number of amino acid substitutions.

Table S9 reports the near-duplicate statistics across all eight families. The results are unambiguous: zero generated sequences across all eight families have $> 95\%$ or $> 90\%$ identity to any stored sequence, and zero sequences are within 2 substitutions of a stored pattern. Only in the Defensin_beta family (L=36, the shortest alignment) do 6% of generated sequences exceed 80% identity to a stored sequence, which corresponds to at most 7 substitutions in a 36-residue sequence. The mean nearest-neighbor identity for generated sequences (0.515–0.655) is substantially higher than the mean within-family pairwise identity (0.217–0.368), indicating that generated sequences are closer to individual stored patterns than stored patterns are to each other on average, but not trivially close. These results confirm that SA generates sequences in intermediate regions of sequence space rather than copying stored patterns.

S13 Wall-Clock Timing Comparison

Table S10 reports wall-clock generation times for 150 sequences across all eight families. SA and profile HMM emit were timed on a single CPU core (Apple M2, 16 GB RAM) with all preprocessing included (PCA construction, phase transition detection, sampling, and decoding for SA; HMM building and sequence emission for HMM emit). EvoDiff and MSA Transformer times were measured on the same hardware (CPU only, no graphics processing unit), and Potts model (plmDCA) times include both parameter fitting and Gibbs sampling.

SA generated 150 sequences in 0.15–4.6 s across all eight families, with the variation driven primarily by the number of stored patterns K (which determines the cost of the softmax attention at each Langevin step). The largest family (WW, $K{=}420$) required 4.6 s, while all families with $K < 100$ completed in under 1 s. Profile HMM emit was consistently the fastest method (0.02–0.09 s), as expected for a method that samples from a pre-computed position-specific model with no iterative computation. EvoDiff required 2–14 hours on CPU depending on sequence length, and the MSA Transformer required 1–8 hours for 50 rounds of iterative masked language model sampling. These comparisons confirm that SA occupies a qualitatively different computational regime from learned baselines: it is 3–5 orders of magnitude faster than EvoDiff and the MSA Transformer, and 2–3 orders of magnitude faster than Potts model fitting, while requiring no GPU and no pretraining.

S14 Consensus-with-Noise Baseline

A natural concern is that SA may function as a "consensus generator" that merely perturbs the family consensus sequence. To test this directly, we implemented a control that adds calibrated noise to the consensus sequence in PCA space: for each family, we computed the position-wise consensus sequence (most frequent amino acid per column), one-hot encoded it, projected it into the same PCA space used by SA, and generated 150 sequences by adding isotropic Gaussian noise with standard deviation matched to the empirical spread of stored patterns in PCA space. We then decoded each noisy PCA vector via inverse PCA and argmax, and evaluated the resulting sequences using the same metrics as SA (Table S11).

The comparison reveals a characteristic signature that distinguishes SA from consensus perturbation. The consensus-with-noise baseline produces sequences with systematically higher novelty (0.63–0.79 vs. 0.40–0.65 for SA) and lower sequence identity to stored patterns (0.44–0.62 vs. 0.52–0.66), reflecting the fact that isotropic noise in PCA space scatters sequences away from all stored patterns rather than interpolating between them. The KL divergence pattern is mixed: the consensus baseline achieves lower KL in some families (e.g., RRM: 0.013 vs. 0.060) but higher KL in others (e.g., zf-C2H2: 0.095 vs. 0.013), showing that isotropic noise can either improve or degrade composition depending on whether the family's covariance aligns with the noise directions. The key distinction is that SA generation maintains higher sequence identity to stored patterns while achieving substantial novelty, indicating that it explores the family's sequence space in a structured way that respects the correlational landscape, whereas consensus perturbation explores isotropically without regard to the family's covariance structure.

S15 Column-Permuted Alignment Control

To test directly whether SA captures information beyond marginal per-position amino acid frequencies, we constructed a control alignment for each family by independently shuffling each column. This procedure preserves the exact per-position amino acid frequency distribution while destroying all inter-position correlations (covariation, structural contacts, epistatic couplings). We then ran the full SA pipeline on the permuted alignment and compared the output against SA run on the real alignment (Table S12).

The results reveal a consistent pattern across all eight families. SA on the real alignment produces sequences with lower novelty but higher sequence identity to stored patterns compared to SA on the permuted alignment. This is expected: the real alignment contains inter-position correlations that constrain the energy landscape and keep generated sequences closer to family-typical regions, whereas the permuted alignment provides only position-independent frequency information, yielding an energy landscape that allows more isotropic exploration. The MI correlation differences between real and permuted SA are small but consistently favor the real alignment in seven of eight families (the exception being zf-C2H2, where the difference is < 0.01), with the magnitude of the difference ranging from 0.01 to 0.08. The larger differences in novelty and sequence identity confirm that the inter-position correlations present in real alignments shape the generated ensemble in ways that marginal frequencies alone cannot reproduce.

S16 PCA Variance Threshold Sensitivity

To assess whether the default 95% variance threshold for PCA dimensionality reduction is a fragile design choice, and to identify the minimum information content needed for reliable generation, we repeated the full SA generation pipeline at eleven retained-variance thresholds spanning 50% to 99%. For each family and threshold, we constructed the memory matrix, identified the critical temperature β^* via entropy inflection, and generated 150 sequences (30 chains $\times$ 5 samples) at $\beta_{\text{gen}} = \max(2\beta^*, 5)$, rounded to the nearest integer. Table S13 reports six representative thresholds for all eight families.

The results reveal two regimes separated by a transition near 70–75% retained variance. Above this threshold, generation quality is robust: KL divergence remained below 0.09 in every condition, novelty broadly increased with the variance threshold (higher d provides more dimensions for exploration), and nearest sequence identity was essentially constant within each family. The critical temperature β^* was largely insensitive to the threshold: in six of eight families, β^* was identical across all eleven thresholds, reflecting the discrete resolution of the entropy inflection search grid. Below 70% retained variance, compositional fidelity began to degrade in the more diverse families. The clearest example was the WW domain (K=420), where KL divergence was 0.03 at 80% but rose to 0.15 at 50%, a five-fold increase indicating that discarded principal components encoded position-specific conservation patterns needed for faithful generation. By contrast, families with lower intrinsic diversity (PDZ, Pkinase, Defensin_beta) were nearly flat across the entire range because their first few principal components already captured the relevant variation. The transition is thus family-dependent, governed by the alignment's spectral structure: families whose eigenvalue spectrum decays slowly require more principal components to preserve compositional fidelity, while compact families tolerate aggressive compression. These findings establish that the 95% default lies well within the robust regime and that practitioners can safely vary the threshold between 75% and 99% without substantially affecting output quality.

S17 Sequence Encoding, PCA Projection, and Decoding

SA operates on continuous vectors on the unit sphere $\mathbb{S}^{d-1}$, while protein sequences are discrete objects over a 20-letter amino acid alphabet. Bridging the two requires three transformations (one-hot encoding, PCA dimensionality reduction, and unit-norm projection), and the reverse path (inverse PCA followed by argmax decoding) recovers discrete sequences from continuous samples. Given a cleaned alignment of K sequences, each of length L amino acids over the standard 20-letter alphabet $\mathcal{A} = \{\text{A}, \text{R}, \text{N}, \ldots, \text{V}\}$, we encode each sequence as a binary vector in $\mathbb{R}^{20L}$ by concatenating per-position indicator vectors:

$$\mathbf{x}_k = \left[\mathbf{e}_{a_1^{(k)}},\ \mathbf{e}_{a_2^{(k)}},\ \ldots,\ \mathbf{e}_{a_L^{(k)}}\right]^\top \in \{0,1\}^{20L}, \tag{S1}$$

where $\mathbf{e}_a \in \{0,1\}^{20}$ is the standard basis vector for amino acid a. The full-dimensional representation is $d_{\text{full}} = 20L$, which ranges from 460 (zf-C2H2, L=23) to $5{,}240$ (Pkinase, L=262) across the eight families studied. This encoding is lossless and treats each position-amino acid pair as an independent coordinate, making no assumptions about residue similarity or physicochemical properties.

The full one-hot space is vastly over-dimensioned relative to the number of stored patterns; the ratio d_{full}/K ranges from 1.5 (WW) to 142 (Pkinase). This creates two problems for SA. First, in the modern Hopfield energy, the similarity scores $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ have variance $1/d$ for a random query on $\mathbb{S}^{d-1}$; when d is large, scores concentrate tightly around zero and extremely high β is required to differentiate among stored patterns, which makes the energy landscape nearly flat and renders Langevin sampling inefficient. Second, the one-hot encoding introduces massive redundancy: at each position, only 1 of 20 coordinates is nonzero. PCA removes this redundancy by projecting onto the directions of actual variation. We compute the economy singular value decomposition of the centered data matrix $\tilde{\mathbf{X}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ and select the smallest p such that the retained variance fraction $\rho(p) = \sum_{j=1}^{p} \sigma_j^2 / \sum_j \sigma_j^2 \geq 0.95$. Among all rank-$p$ linear projections, PCA minimizes the mean squared reconstruction error, making it the optimal linear dimensionality reduction for preserving the structure of the data. Across the eight families, the retained PCA dimension d ranged from 34 (Pkinase) to 186 (WW).

After PCA, we normalize each projected vector to unit L_2 norm: $\mathbf{m}_k = \mathbf{z}_k/\|\mathbf{z}_k\|_2 \in \mathbb{S}^{d-1}$, which ensures that the similarity scores lie in $[-1, 1]$ and that the concentration-of-measure results hold exactly. To decode a Langevin sample $\boldsymbol{\xi} \in \mathbb{R}^d$ back to an amino acid sequence, we apply inverse PCA ($\hat{\mathbf{x}} = \bar{\mathbf{x}} + \mathbf{W}_p\boldsymbol{\xi}$) followed by per-position argmax decoding: for each position ℓ, the amino acid with the largest coordinate in the 20-dimensional sub-vector is selected. This decoding is deterministic and produced 100% valid amino acids across all families and conditions. Several properties make PCA particularly well suited for the modern Hopfield energy: linearity (both projection and inverse are matrix multiplications, keeping the score function in closed form), variance-optimality (it captures maximum family-level variation in the fewest dimensions), and parameter-free determinism (the projection is uniquely defined by the SVD and variance threshold, with no hyperparameters to tune). Algorithm 1 summarizes the complete pipeline.

Algorithm 1 Stochastic Attention Protein Sequence Generation

Require: Seed alignment $\mathbf{S} \in \mathcal{A}^{K\times L}$, variance threshold $\rho_{\min} = 0.95$, step size $\alpha = 0.01$, chains $N_c = 30$, iterations $T = 5000$, burn-in $T_b = 2000$, thinning $\Delta t = 100$, samples/chain $s = 5$
Ensure: Generated sequences $\{\mathbf{g}_1, \dots, \mathbf{g}_S\} \subset \mathcal{A}^L, \quad S = N_c \cdot s$
Encoding:
One-hot encode: $\mathbf{X} \leftarrow [\mathrm{onehot}(\mathbf{S}_{1,:}), \dots, \mathrm{onehot}(\mathbf{S}_{K,:})] \in \{0,1\}^{20L\times K}$
Center: $\bar{\mathbf{x}} \leftarrow \frac{1}{K}\sum_k \mathbf{X}_{:,k}; \quad \tilde{\mathbf{X}} \leftarrow \mathbf{X} - \bar{\mathbf{x}}\mathbf{1}_K^\top$
SVD: $\tilde{\mathbf{X}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$
Select d: smallest p such that $\sum_{j=1}^{p}\sigma_j^2/\sum_j \sigma_j^2 \geq \rho_{\min}$
Project and normalize: $\mathbf{m}_k \leftarrow \mathbf{W}_d^\top(\mathbf{X}_{:,k} - \bar{\mathbf{x}})/\|\mathbf{W}_d^\top(\mathbf{X}_{:,k} - \bar{\mathbf{x}})\|$ for $k = 1, \dots, K$
Assemble memory: $\hat{\mathbf{X}} \leftarrow [\mathbf{m}_1, \dots, \mathbf{m}_K] \in \mathbb{R}^{d\times K}$
Temperature selection:
Find β^*: entropy inflection of $H(\beta) = -\sum_k p_k \log p_k$ with $p_k = \mathrm{softmax}(\beta\,\hat{\mathbf{X}}^\top\mathbf{m}_k)_k$
Set $\beta_{\text{gen}} \leftarrow \max\big(\lceil 2\beta^*\rceil,\ 5\big)$
Multi-chain Langevin sampling:
$\mathcal{G} \leftarrow \emptyset$
for $c = 1, \dots, N_c$ **do**
  Draw $k_c \sim \mathrm{Uniform}\{1, \dots, K\}$
  $\boldsymbol{\xi}_0 \leftarrow \mathbf{m}_{k_c} + 0.01\,\boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$
  for $t = 0, \dots, T-1$ **do**
    $\mathbf{a}_t \leftarrow \mathrm{softmax}(\beta_{\text{gen}}\,\hat{\mathbf{X}}^\top\boldsymbol{\xi}_t)$
    $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$
    $\boldsymbol{\xi}_{t+1} \leftarrow (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\hat{\mathbf{X}}\,\mathbf{a}_t + \sqrt{2\alpha/\beta_{\text{gen}}}\,\boldsymbol{\epsilon}_t$
  end for
  Collect s evenly spaced samples from $\{\boldsymbol{\xi}_{T_b+1}, \boldsymbol{\xi}_{T_b+\Delta t+1}, \dots, \boldsymbol{\xi}_T\}$ into $\mathcal{G}$
end for
Decoding:
for each $\boldsymbol{\xi} \in \mathcal{G}$ **do**
  Inverse PCA: $\hat{\mathbf{x}} \leftarrow \bar{\mathbf{x}} + \mathbf{W}_d\,\boldsymbol{\xi} \in \mathbb{R}^{20L}$
  Argmax decode: $g_\ell \leftarrow \arg\max_{a\in\mathcal{A}}\ \hat{x}_{20(\ell-1)+a}$ for $\ell = 1, \dots, L$
end for
return $\{\mathbf{g}_1, \dots, \mathbf{g}_S\}$

S18 Analysis of the Critical Temperature

We developed an analytical framework to understand the empirical relationship $\beta^* \approx 1.57 + 0.28\sqrt{d}$ reported in Materials and Methods, starting from exact results on similarity variance, building a Gaussian mean-field prediction, and then diagnosing why the mean-field coefficient overshoots and how stored-pattern self-similarity resolves the discrepancy. After one-hot encoding and PCA projection to d dimensions, the stored patterns $\mathbf{m}_k$ are normalized to lie on $\mathbb{S}^{d-1}$. For a random query $\boldsymbol{\xi}$ drawn uniformly from $\mathbb{S}^{d-1}$, each similarity $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ has mean zero and variance $1/d$, regardless of the structure of $\mathbf{m}_k$, a consequence of the rotational invariance of the uniform distribution on the sphere. We verified this numerically for all eight families using 1,000 random probes per family: the ratio $\sigma^2 \cdot d$ is 1.000 ± 0.003 across all families and WW subsamples (Fig. S8B,F), and the excess kurtosis is mildly negative (-0.06 to -0.33), consistent with sub-Gaussian behavior on $\mathbb{S}^{d-1}$. The characteristic similarity scale is therefore $\sigma = 1/\sqrt{d}$, and since β enters the softmax as βe_k, the inflection must occur when $\beta\sigma = \mathcal{O}(1)$, giving $\beta^* \sim \sqrt{d}$.

These observations motivated a Gaussian mean-field model: if the K similarity scores were i.i.d. draws from $\mathcal{N}(0, 1/d)$, the softmax entropy $H(\tau) = -\sum_k p_k \log p_k$ with $p_k \propto \exp(\tau z_k)$ undergoes a transition from $H = \log K$ (uniform, $\tau = 0$) to $H \to 0$ ($\tau \to \infty$). We defined the dimensionless inflection τ^* as the τ that maximizes $|d^2 H/d(\log \tau)^2|$ and computed it numerically by averaging over 500 Gaussian realizations for each $K \in \{10, 20, \dots, 1000\}$. We found that τ^* is nearly constant at ≈ 1.6 for $K \geq 30$, with only weak logarithmic growth (Fig. S8A). Translating back to physical units, the Gaussian prediction $\beta^* = \tau^* \sqrt{d} \approx 1.6\sqrt{d}$ has the correct $\sqrt{d}$ scaling and achieves Pearson $r = 0.98$ with the empirical values, but systematically overpredicts β^* by a factor of ≈ 4.

This overprediction arises because β^* is computed at stored patterns $\mathbf{m}_k$, not at random queries. When $\boldsymbol{\xi} = \mathbf{m}_k$, the self-similarity score $\mathbf{m}_k^\top \mathbf{m}_k = 1$ is an outlier relative to the cross-similarities $\mathbf{m}_j^\top \mathbf{m}_k$ for $j \neq k$, which have magnitudes $\mathcal{O}(1/\sqrt{d})$. The resulting gap $\Delta = 1 - \max_{j \neq k} \mathbf{m}_j^\top \mathbf{m}_k$ ranges from 0.55 to 0.82 across families (Fig. S10E), meaning the softmax concentrates on the self-pattern at much lower β than would be needed for a random query. Substituting the cross-similarity standard deviation σ_{cross} for $1/\sqrt{d}$ fails decisively ($R^2 = -9.0$; Fig. S10D), because the cross-similarity distribution is heavy-tailed (excess kurtosis 6–25) and the self-similarity outlier violates the Gaussian assumption. The fitted model $\beta^* = 1.57 + 0.28\sqrt{d}$ nonetheless remains robust: the $\sqrt{d}$ scaling is set by concentration of measure, and the reduced coefficient 0.28 (vs. $\tau^* \approx 1.6$) encapsulates the combined effect of self-similarity anchoring and the protein family correlation structure.

To quantify uncertainty in the fitted regression, we performed a nonparametric bootstrap analysis: resampling the 33 observations (8 families plus 25 WW scaling replicates) with replacement 10,000 times yielded bootstrap SE of 0.07 for the intercept and 0.007 for the slope, with 95% confidence intervals of $[1.39, 1.68]$ and $[0.266, 0.294]$, respectively. The bootstrap R^2 distribution has a median of 0.969 (95% CI $[0.940, 0.987]$). Because 25 of the 33 data points come from WW subsamples, we also refit the model using only the eight Pfam families: the resulting fit, $\beta^* = 1.64 + 0.278\sqrt{d}$ (R^2=0.95), is nearly identical, confirming that the result is not driven by overrepresentation of a single family. Leave-one-family-out cross-validation achieved R^2=0.96 (root mean square error, RMSE=0.18) on the full dataset and R^2=0.93 (RMSE=0.19) on the 8-family subset, with no individual family acting as an outlier (Table S14).

S19 Regularity and Convergence Properties

The modern Hopfield energy has Lipschitz-continuous gradient with constant $L = 1 + \beta\sigma_{\max}^2/2$, where $\sigma_{\max} = \|\mathbf{X}\|_{\mathrm{op}}$, and satisfies the dissipativity condition $\langle \nabla E(\boldsymbol{\xi}), \boldsymbol{\xi} \rangle \geq \|\boldsymbol{\xi}\|^2/2 - M^2/2$ with $M = \max_k \|\mathbf{m}_k\|$. These regularity properties determine the convergence regime of the Langevin sampler. When $\beta\sigma_{\max}^2 < 2$, the energy is strictly convex and ULA iterates converge to p_β in Wasserstein-2 distance at a geometric rate [5, 6]. In the protein setting, operating temperatures satisfy $\beta\sigma_{\max}^2 \gg 2$, placing the sampler in a non-convex regime where the energy landscape has multiple metastable basins corresponding to stored patterns or clusters thereof. Despite this non-convexity, the multi-chain protocol with long burn-in and thinning yields well-mixed chains (Section S10), and the initialization sensitivity analysis (Section S11) confirms that the sampled distribution is independent of starting conditions.

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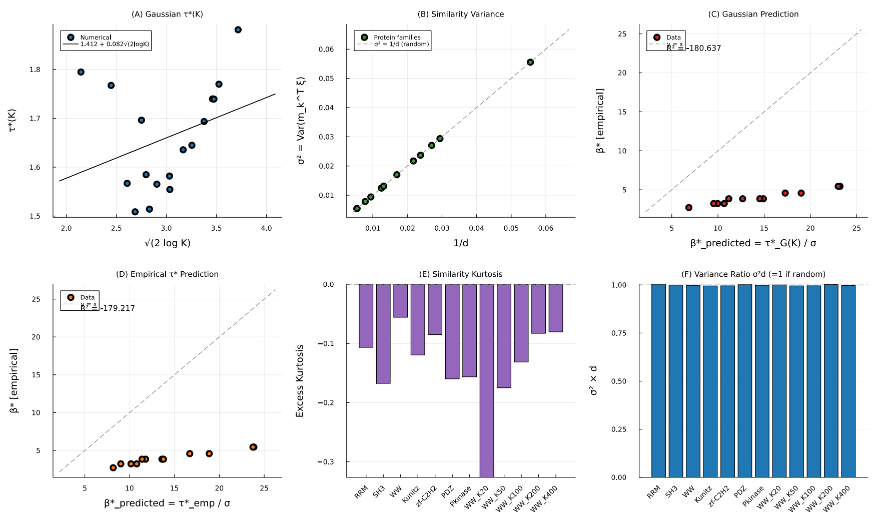

Figure S8: **Analytical investigation of β^*.** (A) Dimensionless softmax inflection $\tau^*(K)$ for Gaussian scores. (B) Similarity variance σ^2 vs. $1/d$ for random probes; all points lie on the identity line. (C) Gaussian prediction $\beta^*_{\text{pred}} = \tau^*_G(K)/\sigma$ vs. empirical β^*; correct trend ($r = 0.98$) but overshoots by $\sim 4\times$. (D) Same as (C) using the empirical τ^*; overshoot persists. (E) Excess kurtosis of the similarity distribution. (F) The ratio $\sigma^2 d$ across all datasets, confirming $\sigma^2 d = 1$.

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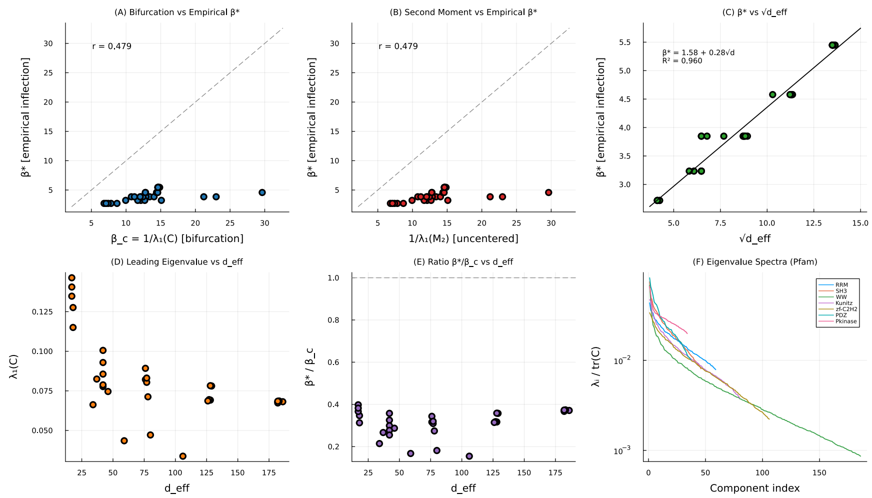

Figure S9: **Bifurcation-based prediction of β^*.** (A) $\beta_c = 1/\lambda_1(\mathbf{C})$ vs. empirical β^* (r=0.479); bifurcation overshoots by 3–5×. (B) Second-moment predictor performance is identical. (C) β^* vs. $\sqrt{d}$ with fitted model (R^2=0.960). (D) Leading eigenvalue vs. d; no consistent relationship. (E) Ratio β^*/β_c confirms bifurcation systematically overestimates. (F) Normalized eigenvalue spectra for all eight families.

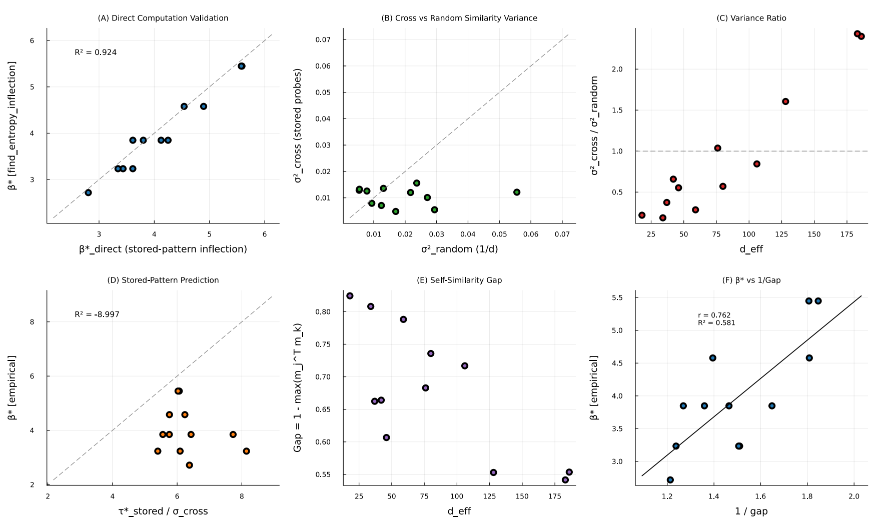

Figure S10: **Stored-pattern probe analysis.** (A) β^* from stored-pattern inflection vs. the value from the entropy inflection search. (B) Cross-similarity variance (stored probes) vs. random-probe variance. (C) Variance ratio vs. d. (D) The decomposition $\tau^*_{\text{stored}}/\sigma_{\text{cross}}$ fails as a predictor ($R^2 = -9.0$). (E) Self-similarity gap Δ vs. d. (F) β^* vs. $1/\Delta$ ($r = 0.76$).

Table S1: Generation metrics across eight Pfam families (mean $\pm$ SE). SA gen: generation regime ($\beta \approx 2\beta^*$). SA ret: retrieval regime ($\beta \approx 20\beta^*$). BS: bootstrap. GP: Gaussian perturbation. CC: convex combination. KL: amino acid composition KL divergence (↓ lower is better; SE via 1,000-fold bootstrap). Novelty: $1 - \max_k \cos(\hat{\boldsymbol{\xi}}, \mathbf{m}_k)$ in PCA space (↑ higher is better). SeqID: nearest sequence identity to a stored pattern after decoding (↓ lower is better). Bold: best method per family and metric.

Family	Method	$\mathrm{KL}_{\mathrm{AA}}$ (↓)	Novelty (↑)	SeqID (↓)
RRM	SA gen	$\mathbf{0.060 \pm 0.005}$	$\mathbf{0.513 \pm 0.011}$	$\mathbf{0.538 \pm 0.006}$
	SA ret	0.107 ± 0.010	0.334 ± 0.020	0.616 ± 0.012
	BS	0.133 ± 0.006	0.000 ± 0.000	0.645 ± 0.004
	GP	0.132 ± 0.005	0.008 ± 0.000	0.633 ± 0.006
	CC	2.091 ± 0.000	0.510 ± 0.009	0.562 ± 0.000
SH3	SA gen	0.036 ± 0.004	$\mathbf{0.471 \pm 0.012}$	$\mathbf{0.593 \pm 0.006}$
	SA ret	0.035 ± 0.004	0.213 ± 0.012	0.722 ± 0.011
	BS	$\mathbf{0.030 \pm 0.003}$	0.000 ± 0.000	0.772 ± 0.006
	GP	$\mathbf{0.030 \pm 0.003}$	0.006 ± 0.000	0.752 ± 0.006
	CC	0.798 ± 0.012	0.474 ± 0.008	0.601 ± 0.002
WW	SA gen	$\mathbf{0.008 \pm 0.002}$	$\mathbf{0.653 \pm 0.006}$	$\mathbf{0.562 \pm 0.005}$
	SA ret	0.046 ± 0.013	0.354 ± 0.016	0.763 ± 0.011
	BS	0.024 ± 0.003	0.000 ± 0.000	0.823 ± 0.005
	GP	0.029 ± 0.004	0.017 ± 0.000	0.809 ± 0.006
	CC	1.375 ± 0.092	0.637 ± 0.004	0.766 ± 0.001
Kunitz	SA gen	$\mathbf{0.013 \pm 0.002}$	$\mathbf{0.557 \pm 0.009}$	$\mathbf{0.607 \pm 0.004}$
	SA ret	0.037 ± 0.003	0.257 ± 0.016	0.760 ± 0.012
	BS	0.023 ± 0.002	0.000 ± 0.000	0.834 ± 0.004
	GP	0.024 ± 0.002	0.010 ± 0.000	0.844 ± 0.004
	CC	0.721 ± 0.013	0.546 ± 0.007	0.590 ± 0.001
zf-C2H2	SA gen	0.013 ± 0.028	$\mathbf{0.596 \pm 0.010}$	$\mathbf{0.557 \pm 0.007}$
	SA ret	0.021 ± 0.010	0.167 ± 0.014	0.848 ± 0.013
	BS	$\mathbf{0.007 \pm 0.002}$	0.000 ± 0.000	0.938 ± 0.004
	GP	$\mathbf{0.007 \pm 0.002}$	0.011 ± 0.000	0.940 ± 0.004
	CC	2.346 ± 0.060	0.580 ± 0.006	0.600 ± 0.001
PDZ	SA gen	$\mathbf{0.038 \pm 0.010}$	$\mathbf{0.418 \pm 0.009}$	$\mathbf{0.543 \pm 0.005}$
	SA ret	0.059 ± 0.016	0.236 ± 0.017	0.626 ± 0.014
	BS	0.041 ± 0.002	0.000 ± 0.000	0.642 ± 0.009
	GP	0.045 ± 0.003	0.006 ± 0.000	0.648 ± 0.007
	CC	0.760 ± 0.030	0.443 ± 0.010	0.523 ± 0.003
Pkinase	SA gen	$\mathbf{0.044 \pm 0.003}$	$\mathbf{0.397 \pm 0.010}$	$\mathbf{0.514 \pm 0.003}$
	SA ret	0.067 ± 0.010	0.173 ± 0.017	0.535 ± 0.012
	BS	0.078 ± 0.005	0.000 ± 0.000	0.547 ± 0.007
	GP	0.069 ± 0.005	0.006 ± 0.001	0.540 ± 0.007
	CC	1.640 ± 0.030	0.364 ± 0.008	0.531 ± 0.001
Def._beta	SA gen	$\mathbf{0.024 \pm 0.005}$	$\mathbf{0.400 \pm 0.012}$	$\mathbf{0.656 \pm 0.007}$
	SA ret	0.030 ± 0.007	0.190 ± 0.016	0.788 ± 0.013
	BS	0.038 ± 0.003	0.000 ± 0.000	0.780 ± 0.007
	GP	0.035 ± 0.003	0.009 ± 0.001	0.768 ± 0.006
	CC	0.590 ± 0.020	0.390 ± 0.010	0.638 ± 0.002

Table S2: SA generation quality at $K=20$ across four Pfam families (five independent subsamples each). Values are mean $\pm$ SD over the five replicates.

Family	K_{full}	L	d_{PCA}	KL	Novelty	Seq. ID
SH3	55	48	17–18	0.026 ± 0.003	0.46 ± 0.01	0.67 ± 0.01
Kunitz	99	53	17–18	0.020 ± 0.004	0.47 ± 0.01	0.72 ± 0.01
zf-C2H2	151	23	18	0.013 ± 0.004	0.47 ± 0.01	0.73 ± 0.00
WW	420	31	17–18	0.019 ± 0.008	0.47 ± 0.01	0.71 ± 0.02

Table S3: Pairwise mutual information (MI) preservation: Pearson correlation r between upper-triangle MI vectors of stored sequences and sequences generated by each method. Higher values indicate better preservation of pairwise residue covariation. Potts model results are shown only for the two families where plmDCA sequences were generated.

Family	SA gen	SA ret	Potts	HMM	EvoDiff	MSAT
RRM	0.53	0.34	N/A	0.49	0.86	0.60
SH3	0.66	0.64	0.75	0.46	0.88	0.73
WW	0.71	0.49	N/A	0.29	0.83	0.58
Kunitz	0.80	0.69	0.89	0.32	0.87	0.50
zf-C2H2	0.92	0.87	N/A	0.72	0.97	0.92
PDZ	0.59	0.44	N/A	0.07	0.81	0.01
Pkinase	0.05	-0.09	N/A	0.23	0.59	0.05
Defensin_beta	0.85	0.75	N/A	0.83	0.95	0.88

Table S4: ESM2-650M pseudo-perplexity validation. Lower pseudo-perplexity indicates a more protein-like sequence. p: Wilcoxon rank-sum test (two-sided). d: Cohen's effect size (negative values indicate SA is better). Significance: $^{***}p < 0.001$, $^{**}p < 0.01$, n.s. = not significant ($p > 0.05$).

Family	SA gen ppl	Stored ppl	p	d	Sig.
RRM	4.87 ± 0.68	5.20 ± 1.92	0.30	-0.29	n.s.
SH3	5.24 ± 0.82	4.62 ± 1.48	< 0.001	0.60	***
WW	6.78 ± 1.77	5.04 ± 1.36	< 0.001	1.03	***
Kunitz	4.47 ± 0.57	4.98 ± 0.94	< 0.001	-0.74	***
zf-C2H2	7.26 ± 1.66	8.10 ± 3.47	0.41	-0.37	n.s.
PDZ	6.30 ± 1.21	5.31 ± 2.61	0.002	0.60	**
Pkinase	4.44 ± 0.32	4.32 ± 0.80	0.39	0.27	n.s.
Defensin_beta	6.35 ± 1.27	6.82 ± 2.09	0.32	-0.31	n.s.

Table S5: Deep mutational scanning validation. SA-generated substitutions relative to the family consensus were matched to published single-mutant DMS fitness data. Tolerance rate: fraction of matched substitutions classified as functional. DMS overall: baseline tolerance rate across all single mutations in the dataset. Enrichment: ratio of SA tolerance rate to DMS overall rate.

Family	DMS source	Matched	SA tol. rate	DMS overall	Enrichment	p
PDZ	McLaughlin et al. 2012	4,313	91.7%	77.5%	$1.18\times$	$< 10^{-135}$
SH3	Faure et al. 2022	2,396	90.4%	67.5%	$1.34\times$	$< 10^{-156}$

Table S6: Potts model (plmDCA) baseline comparison for two Pfam families. Metrics are computed identically to the main baseline analysis (30 chains $\times$ 5 samples). Data/param: ratio of training sequences to model parameters.

Family	Method	$\mathrm{KL}_{\mathrm{AA}}$ ($\downarrow$)	Novelty ($\uparrow$)	SeqID	Data/param
SH3	SA generation	0.036 ± 0.026	0.471 ± 0.012	0.593 ± 0.007	N/A
	Potts (plmDCA)	0.025 ± 0.019	0.374 ± 0.011	0.539 ± 0.009	1.1×10^{-4}
	HMM emit	0.009 ± 0.017	0.614 ± 0.004	0.290 ± 0.006	N/A
Kunitz	SA generation	0.013 ± 0.022	0.557 ± 0.009	0.607 ± 0.004	N/A
	Potts (plmDCA)	0.010 ± 0.017	0.540 ± 0.010	0.548 ± 0.010	1.6×10^{-4}
	HMM emit	0.030 ± 0.011	0.666 ± 0.004	0.229 ± 0.007	N/A

Table S7: Sampling diagnostics for ULA and MALA chains across eight Pfam families. τ_{int}: integrated autocorrelation time of the Hopfield energy (mean over 10 chains). ESS: effective sample size per chain (n=3,000 post-burn-in samples). MALA AR: Metropolis-adjusted acceptance rate. Conv. iter: mean iteration at which burn-in energy converged (1% criterion). Margin: ratio of burn-in budget (2,000) to convergence iteration.

Family	$\tau_{\mathrm{int}}^{\mathrm{ULA}}$	$\mathrm{ESS}^{\mathrm{ULA}}$	$\tau_{\mathrm{int}}^{\mathrm{MALA}}$	$\mathrm{ESS}^{\mathrm{MALA}}$	MALA AR	Conv. iter	Margin
RRM	76	42	99	35	0.998	509	$3.9\times$
SH3	95	35	84	39	0.998	520	$3.8\times$
WW	86	37	96	34	0.996	369	$5.4\times$
Kunitz	100	35	109	33	0.997	500	$4.0\times$
zf-C2H2	105	33	82	39	0.997	447	$4.5\times$
PDZ	113	32	131	27	0.998	304	$6.6\times$
Pkinase	83	38	98	36	0.998	617	$3.2\times$
Defensin_beta	122	32	124	29	0.998	306	$6.5\times$

Table S8: Initialization sensitivity: stored-pattern vs. random sphere initialization. All metrics are computed identically to the main analysis (30 chains $\times$ 5 samples). The two strategies produce indistinguishable output across all eight families.

Family	Init	$\mathrm{KL}_{\mathrm{AA}}$	Novelty	SeqID	MaxCos
RRM	Stored	0.060	0.513	0.538	0.487
	Random	0.059	0.511	0.539	0.489
SH3	Stored	0.036	0.471	0.593	0.529
	Random	0.036	0.471	0.593	0.529
WW	Stored	0.008	0.653	0.562	0.347
	Random	0.008	0.653	0.562	0.347
Kunitz	Stored	0.013	0.557	0.607	0.443
	Random	0.013	0.557	0.607	0.443
zf-C2H2	Stored	0.013	0.596	0.557	0.404
	Random	0.013	0.596	0.557	0.404
PDZ	Stored	0.038	0.418	0.543	0.582
	Random	0.038	0.418	0.543	0.582
Pkinase	Stored	0.035	0.478	0.515	0.522
	Random	0.035	0.478	0.515	0.522
Defensin_beta	Stored	0.022	0.402	0.655	0.598
	Random	0.022	0.402	0.655	0.598

Table S9: Near-duplicate analysis: sequence identity between SA-generated and stored sequences. MaxID: mean maximum identity of each generated sequence to its nearest stored neighbor. %>95, %>90, %>80: fraction of generated sequences exceeding the indicated identity threshold. %≤2sub: fraction within 2 substitutions of a stored sequence. Stored ID: mean pairwise identity among stored sequences.

Family	Mean MaxID	%>95	%>90	%>80	%≤2sub	Stored ID
RRM	0.538	0.0	0.0	0.0	0.0	0.230
SH3	0.593	0.0	0.0	0.7	0.0	0.288
WW	0.562	0.0	0.0	0.0	0.0	0.324
Kunitz	0.607	0.0	0.0	0.0	0.0	0.368
zf-C2H2	0.557	0.0	0.0	0.0	0.0	0.304
PDZ	0.543	0.0	0.0	0.7	0.0	0.217
Pkinase	0.515	0.0	0.0	0.0	0.0	0.274
Defensin_beta	0.655	0.0	0.0	6.0	0.0	0.332

Table S10: Wall-clock time for generating 150 sequences per family. SA and HMM times are averaged over three runs on a single CPU core (Apple M2). EvoDiff and MSA Transformer times are from CPU-only runs. Potts times include pseudo-likelihood fitting and Gibbs sampling.

Family	L	K	SA (s)	HMM (s)	EvoDiff (h)	MSAT (h)	Potts (min)
RRM	71	68	0.5	0.05	~ 3	~ 4	N/A
SH3	48	55	0.4	0.03	~ 2	~ 2	~ 5
WW	31	420	4.6	0.03	~ 2	~ 3	N/A
Kunitz	53	99	0.7	0.03	~ 2	~ 3	~ 5
zf-C2H2	23	151	1.0	0.02	~ 2	~ 1	N/A
PDZ	83	44	0.3	0.04	~ 4	~ 4	N/A
Pkinase	262	37	0.2	0.09	~ 14	~ 8	N/A
Defensin_beta	36	45	0.3	0.03	~ 2	~ 1	N/A

Table S11: Consensus-with-noise baseline vs. SA generation. CN: consensus sequence with calibrated Gaussian noise in PCA space. SA: stochastic attention generation ($\beta \approx 2\beta^*$). Metrics are computed identically to the main analysis.

Family	Method	KL_{AA}	Novelty	SeqID
RRM	CN	0.013 ± 0.002	0.686 ± 0.005	0.487 ± 0.005
	SA gen	0.060 ± 0.005	0.513 ± 0.011	0.538 ± 0.006
SH3	CN	0.004 ± 0.001	0.658 ± 0.005	0.537 ± 0.005
	SA gen	0.036 ± 0.004	0.471 ± 0.012	0.593 ± 0.006
WW	CN	0.021 ± 0.003	0.787 ± 0.002	0.437 ± 0.003
	SA gen	0.008 ± 0.004	0.653 ± 0.006	0.562 ± 0.005
Kunitz	CN	0.004 ± 0.001	0.722 ± 0.004	0.551 ± 0.003
	SA gen	0.013 ± 0.002	0.557 ± 0.009	0.607 ± 0.004
zf-C2H2	CN	0.095 ± 0.002	0.742 ± 0.003	0.495 ± 0.004
	SA gen	0.013 ± 0.027	0.596 ± 0.010	0.557 ± 0.007
PDZ	CN	0.021 ± 0.008	0.633 ± 0.005	0.522 ± 0.005
	SA gen	0.038 ± 0.010	0.418 ± 0.009	0.543 ± 0.005
Pkinase	CN	0.015 ± 0.001	0.630 ± 0.005	0.528 ± 0.004
	SA gen	0.035 ± 0.001	0.478 ± 0.011	0.515 ± 0.004
Defensin_beta	CN	0.013 ± 0.002	0.631 ± 0.005	0.620 ± 0.006
	SA gen	0.022 ± 0.003	0.402 ± 0.013	0.655 ± 0.007

Table S12: Column-permuted alignment control. SA (real): stochastic attention on the original alignment. SA (perm): stochastic attention on a column-permuted alignment that preserves per-position frequencies but destroys inter-position correlations. MI r: Pearson correlation between pairwise MI vectors of stored and generated sequences.

Family	Method	$\mathrm{KL}_{\mathrm{AA}}$	Novelty	SeqID	MI r
RRM	SA (real)	0.060 ± 0.005	0.513 ± 0.011	0.538 ± 0.006	0.53
	SA (perm)	0.050 ± 0.017	0.629 ± 0.005	0.460 ± 0.003	0.48
SH3	SA (real)	0.036 ± 0.004	0.471 ± 0.012	0.593 ± 0.006	0.67
	SA (perm)	0.032 ± 0.004	0.611 ± 0.005	0.500 ± 0.003	0.59
WW	SA (real)	0.008 ± 0.002	0.653 ± 0.006	0.562 ± 0.005	0.73
	SA (perm)	0.006 ± 0.002	0.688 ± 0.004	0.540 ± 0.004	0.73
Kunitz	SA (real)	0.013 ± 0.002	0.557 ± 0.009	0.607 ± 0.004	0.80
	SA (perm)	0.012 ± 0.002	0.660 ± 0.007	0.546 ± 0.003	0.78
zf-C2H2	SA (real)	0.013 ± 0.027	0.596 ± 0.010	0.557 ± 0.007	0.92
	SA (perm)	0.006 ± 0.005	0.654 ± 0.006	0.511 ± 0.004	0.93
PDZ	SA (real)	0.038 ± 0.011	0.418 ± 0.009	0.543 ± 0.005	0.58
	SA (perm)	0.028 ± 0.005	0.545 ± 0.007	0.433 ± 0.003	0.52
Pkinase	SA (real)	0.035 ± 0.001	0.478 ± 0.011	0.515 ± 0.004	0.06
	SA (perm)	0.035 ± 0.001	0.612 ± 0.005	0.449 ± 0.001	0.00
Defensin_beta	SA (real)	0.022 ± 0.003	0.402 ± 0.013	0.655 ± 0.007	0.85
	SA (perm)	0.017 ± 0.002	0.542 ± 0.008	0.552 ± 0.004	0.82

Table S13: PCA variance threshold sensitivity study. For each family and retained-variance threshold, the table reports the PCA dimension d, empirical critical temperature β^*, generation temperature β_{gen}, amino acid composition KL divergence, PCA-space cosine novelty, and nearest sequence identity. The horizontal rule within each family separates the degraded regime ($\leq 70\%$) from the robust regime ($\geq 80\%$).

Family	Threshold	d	β^*	β_{gen}	$\mathrm{KL}_{\mathrm{AA}}$	Novelty	SeqID
RRM	50%	21	3.85	8	0.186	0.392	0.539
	60%	27	3.85	8	0.111	0.426	0.546
	70%	34	3.85	8	0.135	0.463	0.539
	80%	43	3.85	8	0.085	0.497	0.535
	90%	53	3.85	8	0.067	0.508	0.540
	95%	59	3.85	8	0.060	0.513	0.538
SH3	50%	15	3.23	6	0.087	0.312	0.601
	60%	19	3.23	6	0.072	0.372	0.593
	70%	25	3.23	6	0.039	0.402	0.592
	80%	32	3.23	6	0.038	0.447	0.587
	90%	41	3.85	8	0.032	0.444	0.596
	95%	46	3.85	8	0.036	0.471	0.593
WW	50%	33	5.45	11	0.152	0.410	0.646
	60%	47	5.45	11	0.133	0.476	0.631
	70%	67	5.45	11	0.125	0.530	0.624
	80%	95	5.45	11	0.028	0.588	0.600
	90%	142	5.45	11	0.020	0.636	0.576
	95%	186	5.45	11	0.008	0.653	0.562
Kunitz	50%	23	3.85	8	0.053	0.387	0.638
	60%	31	3.85	8	0.047	0.433	0.636
	70%	40	3.85	8	0.031	0.469	0.628
	80%	53	3.85	8	0.022	0.500	0.619
	90%	69	3.85	8	0.023	0.535	0.616
	95%	80	3.85	8	0.013	0.557	0.607
zf-C2H2	50%	28	4.58	9	0.112	0.397	0.610
	60%	38	4.58	9	0.037	0.458	0.599
	70%	49	4.58	9	0.106	0.483	0.595
	80%	65	4.58	9	0.017	0.528	0.583
	90%	88	4.58	9	0.010	0.565	0.572
	95%	106	4.58	9	0.013	0.596	0.557
PDZ	50%	12	2.70	5	0.038	0.269	0.558
	60%	16	3.23	6	0.035	0.305	0.564
	70%	20	3.23	6	0.042	0.337	0.553
	80%	25	3.23	6	0.035	0.359	0.549
	90%	32	3.23	6	0.046	0.400	0.544
	95%	37	3.23	6	0.038	0.418	0.543
Pkinase	50%	15	3.23	6	0.041	0.356	0.501
	60%	18	3.23	6	0.041	0.390	0.500
	70%	22	3.23	6	0.038	0.426	0.506
	80%	27	3.23	6	0.035	0.463	0.506
	90%	31	3.23	6	0.035	0.480	0.509
	95%	34	3.23	6	0.035	0.478	0.515
Defensin_beta	50%	11	2.70	5	0.036	0.223	0.679
	60%	15	3.23	6	0.035	0.262	0.678
	70%	19	3.23	6	0.030	0.307	0.668
	80%	25	3.23	6	0.029	0.312	0.680
	90%	32	3.23	6	0.020	0.381	0.657
	95%	37	3.23	6	0.022	0.402	0.655

Table S14: Leave-one-family-out cross-validation of β^* prediction. Scheme 1: full 33-point dataset. Scheme 2: 8-family-only LOO. Both schemes confirm out-of-sample generalization.

Scheme	n	LOOCV R^2	LOOCV RMSE
Full 33-pt LOFO	33	0.959	0.184
8-family LOO	8	0.928	0.195